

1. A company is considering implementing a BYOD policy. What is the most significant security risk to consider?
 - A. Increased hardware costs
 - B. Reduced network performance
 - C. Possible data leakage
 - D. Potential for decreased productivity

2. A security analyst observes a huge amount of ICMP echo reply (ping) traffic coming from multiple sources to a single destination within the network. What type of attack is most likely occurring?
 - A. Man-in-the-Middle attack
 - B. Smurf attack
 - C. Fraggle attack
 - D. Ping of Death attack

3. Which protocol is vulnerable to sniffing attacks because it transmits usernames and passwords in cleartext?
 - A. SSH
 - B. FTP
 - C. HTTPS
 - D. SNMP

4. A company has a policy that requires all mobile devices to be encrypted. What is the primary purpose of this policy?
 - A. To prevent data loss if a device is physically lost or stolen
 - B. To ensure the devices cannot be used for personal tasks
 - C. To prevent unauthorized access to the corporate Wi-Fi network
 - D. To ensure all software on the devices is up to date

5. A cybersecurity analyst is reviewing a series of failed login attempts on the company's user portal. The attempts appear to be using common usernames and passwords. Which type of attack is most likely being attempted?
 - A. DDoS attack
 - B. Phishing attack
 - C. Brute force attack
 - D. Watering hole attack

6. Which of the following symmetric encryption algorithms is considered to be the most secure?
 - A. DES
 - B. RC4
 - C. AES
 - D. 3DES

7. What does a rootkit primarily target?
- A. Network traffic
 - B. User data
 - C. Operating system
 - D. Firewall rules
8. A cybersecurity analyst has discovered that an attacker has been moving laterally within the network. What is the BEST next step?
- A. Implement a honeypot
 - B. Disconnect the entire network
 - C. Perform a risk assessment
 - D. Contain the compromise
9. Which of the following is considered an example of a technical control?
- A. Security policy
 - B. User training
 - C. Firewall
 - D. Background checks
10. A company has implemented a system to centralize the management of user credentials. What is this system known as?
- A. Network Access Control (NAC)
 - B. Security Information and Event Management (SIEM)
 - C. Identity and Access Management (IAM)
 - D. Data Loss Prevention (DLP)
11. What is the primary purpose of a Web Application Firewall (WAF)?
- A. Block malicious IP addresses
 - B. Protect against SQL injection and XSS attacks
 - C. Filter out spam emails
 - D. Provide VPN access for remote users
12. What is the main difference between a worm and a virus?
- A. A worm is always malicious, while a virus can be benign
 - B. A virus requires user action to spread, while a worm can spread by itself
 - C. A worm can be programmed, while a virus cannot
 - D. A virus targets the operating system, while a worm targets applications

13. Which of the following is a type of public key infrastructure (PKI) attack where the attacker redirects the user to a malicious website that appears to be legitimate?
- A. Replay attack
 - B. Pharming attack
 - C. Session hijacking
 - D. Man-in-the-Middle attack
14. A system administrator finds a file on a user's desktop that appears to contain a list of passwords for various company systems. What type of attack has likely occurred?
- A. Password spraying
 - B. Credential harvesting
 - C. Rainbow table attack
 - D. Birthday attack
15. A company has implemented a system that uses a single secure private key to encrypt and decrypt messages. What type of encryption system is this?
- A. Asymmetric encryption
 - B. Symmetric encryption
 - C. Hash function
 - D. Digital signature
16. A security analyst is examining logs and notices a large number of HTTP GET and POST requests from an IP address that doesn't belong to the company. What type of attack is likely occurring?
- A. DNS poisoning
 - B. ARP spoofing
 - C. SQL injection
 - D. Directory traversal
17. A DDoS attack is overwhelming the company's website. Which of the following would be the most effective way to mitigate this attack?
- A. Install a more powerful server
 - B. Increase the bandwidth of the Internet connection
 - C. Implement rate limiting
 - D. Block all incoming traffic

18. What is the primary purpose of a VLAN?
- A. To increase the speed of the network
 - B. To separate network traffic
 - C. To add additional devices to the network
 - D. To secure wireless connections
19. Which of the following is a risk associated with cloud computing?
- A. Increased cost
 - B. Limited scalability
 - C. Vendor lock-in
 - D. Lack of control over physical security
20. An employee receives an email that appears to be from the CEO asking for sensitive company information. What type of attack is this an example of?
- A. Phishing
 - B. Vishing
 - C. Whaling
 - D. Spear phishing
21. Which form of authentication is based on something the user has?
- A. Password
 - B. Fingerprint
 - C. Security token
 - D. PIN
22. Which of the following is the most critical step in responding to a security incident?
- A. Reporting the incident to law enforcement
 - B. Notifying customers of the incident
 - C. Documenting the incident
 - D. Identifying the root cause of the incident
23. Which of the following is NOT a characteristic of a zero-day vulnerability?
- A. It is unknown to the software vendor
 - B. There is no patch available
 - C. It is often exploited by hackers
 - D. It is easy to detect

24. Which technology is used to separate a physical server into multiple virtual servers?
- A. VPN
 - B. VLAN
 - C. Hypervisor
 - D. Proxy server
25. Which of the following is a security advantage of using a virtual private network (VPN)?
- A. It increases network speed
 - B. It encrypts data in transit
 - C. It prevents DDoS attacks
 - D. It eliminates the need for firewalls
26. A security analyst has detected an anomaly in network traffic. Which tool would be best suited to further investigate this issue?
- A. Network scanner
 - B. Protocol analyzer
 - C. Vulnerability scanner
 - D. Honeypot
27. Which type of malware requires a host program to spread?
- A. Worm
 - B. Trojan
 - C. Rootkit
 - D. Ransomware
28. What is the main benefit of a stateful firewall over a stateless firewall?
- A. It can block malicious IP addresses
 - B. It can filter traffic based on the state of the connection
 - C. It can prevent DDoS attacks
 - D. It can provide VPN access for remote users
29. A company is considering outsourcing its email services to a cloud provider. What type of cloud service model is it considering?
- A. Infrastructure as a Service (IaaS)
 - B. Software as a Service (SaaS)
 - C. Platform as a Service (PaaS)
 - D. Security as a Service (SECaaS)

30. Which type of security testing involves the tester having full knowledge of the system being tested?
- A. Black box testing
 - B. Gray box testing
 - C. White box testing
 - D. Red team testing
31. A company has implemented a policy that requires all employees to use a smart card and a PIN to access the corporate network. What type of authentication method is this?
- A. Biometric authentication
 - B. Single-factor authentication
 - C. Two-factor authentication
 - D. Multifactor authentication
32. Which of the following is a characteristic of a rainbow table attack?
- A. It uses precomputed hashes to crack passwords
 - B. It uses brute force to guess passwords
 - C. It uses social engineering to obtain passwords
 - D. It uses malware to steal passwords
33. What is the primary purpose of a security incident response plan?
- A. To prevent security incidents
 - B. To determine who is responsible for a security incident
 - C. To define the steps to take in response to a security incident
 - D. To document past security incidents
34. What is the primary purpose of a Certificate Authority (CA) in a Public Key Infrastructure (PKI)?
- A. To distribute private keys
 - B. To verify user identities
 - C. To issue digital certificates
 - D. To decrypt messages
35. A security analyst is investigating a potential security incident and notices a large amount of data being transferred from a company server to an unknown IP address. What type of threat is most likely being encountered?
- A. DDoS attack
 - B. Data exfiltration
 - C. Phishing attack
 - D. Malware infection

36. Which of the following is a characteristic of a full backup?
- A. It backs up only the data that has changed since the last backup
 - B. It backs up all data and marks each file as being backed up
 - C. It backs up all data but does not mark each file as being backed up
 - D. It backs up only the system files
37. A company is developing a new web application and wants to ensure it is secure. Which of the following would be the BEST approach?
- A. Conduct a vulnerability scan
 - B. Implement a Web Application Firewall (WAF)
 - C. Conduct a penetration test
 - D. Implement SSL/TLS
38. A company has suffered a data breach and the investigation revealed that an insider was responsible. What type of threat does this represent?
- A. Phishing
 - B. DDoS
 - C. Man-in-the-Middle
 - D. Insider threat
39. Which of the following is a security disadvantage of using peer-to-peer (P2P) networks?
- A. They are susceptible to man-in-the-middle attacks
 - B. They can be used to distribute malware
 - C. They do not support file sharing
 - D. They require a central server
40. Which of the following is a critical step in the patch management process?
- A. Deploy patches immediately upon release
 - B. Test patches before deployment
 - C. Disable automatic updates
 - D. Only apply patches to critical systems
41. Which of the following is an example of a physical security control?
- A. Firewall
 - B. User training
 - C. Security policy
 - D. CCTV cameras

42. A security analyst is conducting a penetration test and has gained access to a system. The analyst now wants to ensure they maintain access to the system, even if the system is rebooted. What technique should the analyst use?
- A. Escalating privileges
 - B. Implementing a backdoor
 - C. Covering tracks
 - D. Social engineering
43. Which of the following is a benefit of using a Host-based Intrusion Detection System (HIDS) over a Network-based Intrusion Detection System (NIDS)?
- A. It can detect attacks that a NIDS cannot, such as attacks that occur within encrypted traffic
 - B. It requires less computational resources to operate
 - C. It can monitor network traffic in real-time
 - D. It can detect attacks coming from outside the network
44. A company is planning to implement a system that will require users to provide two forms of identification from different categories (something they know, something they have, something they are). What type of authentication is this?
- A. Single-factor authentication
 - B. Two-factor authentication
 - C. Multifactor authentication
 - D. Biometric authentication
45. Which of the following is the most secure method for securely disposing of SSD drives?
- A. Degaussing
 - B. Formatting
 - C. Physical destruction
 - D. Overwriting
46. A security analyst has detected a number of failed login attempts on a server from a single IP address. What type of attack is most likely being attempted?
- A. Brute force attack
 - B. DDoS attack
 - C. Man-in-the-middle attack
 - D. Phishing attack

47. A user reports that their computer has been running slowly and they have been receiving pop-up ads while browsing the internet. What type of malware is most likely causing these symptoms?
- A. Ransomware
 - B. Trojan
 - C. Adware
 - D. Rootkit
48. Which of the following is an advantage of using a Security Information and Event Management (SIEM) system?
- A. It provides real-time analysis of security alerts
 - B. It prevents malware infections
 - C. It implements firewall rules
 - D. It manages user passwords
49. A company uses a secure protocol for transferring files between systems. This protocol also allows for management and manipulation of directories on the remote system. What protocol is the company using?
- A. FTP
 - B. SFTP
 - C. TFTP
 - D. SCP
50. Which of the following is a key concept of the CIA triad in cybersecurity?
- A. Non-repudiation
 - B. Integrity
 - C. Availability
 - D. All of the above
51. A security analyst is considering implementing a system that uses behavior-based monitoring to detect threats. What type of system is this?
- A. Intrusion Detection System (IDS)
 - B. Intrusion Prevention System (IPS)
 - C. Network-based Intrusion Detection System (NIDS)
 - D. Host-based Intrusion Detection System (HIDS)
52. A company's web server has been defaced by a hacker. What type of attack has likely occurred?
- A. DDoS attack
 - B. Phishing attack
 - C. Web application attack
 - D. Man-in-the-Middle attack

53. A company wants to implement an authentication system that uses a centralized server to authenticate users. Which of the following would be the best choice?
- A. TACACS+
 - B. CHAP
 - C. MS-CHAP
 - D. PAP
54. Which of the following is a vulnerability in the WEP protocol that makes it insecure?
- A. It does not encrypt data
 - B. It uses static encryption keys
 - C. It is susceptible to man-in-the-middle attacks
 - D. It does not support two-factor authentication
55. Which of the following is the primary purpose of using a hash function in cybersecurity?
- A. To encrypt data
 - B. To verify data integrity
 - C. To compress data
 - D. To authenticate users
56. Which of the following is a characteristic of a stateful inspection firewall?
- A. It can only filter incoming traffic
 - B. It cannot filter traffic based on ports
 - C. It keeps track of the state of network connections
 - D. It cannot block specific IP addresses
57. Which of the following is a type of security control that is implemented after a security incident has occurred?
- A. Preventive control
 - B. Detective control
 - C. Corrective control
 - D. Deterrent control
58. Which of the following is the best definition of non-repudiation in the context of cybersecurity?
- A. The ability to verify the integrity of data
 - B. The ability to ensure the confidentiality of data
 - C. The ability to prevent the denial of actions performed
 - D. The ability to authenticate users

59. A company wants to implement a system that can detect and prevent attacks in real-time. Which of the following would be the best choice?
- A. Intrusion Detection System (IDS)
 - B. Intrusion Prevention System (IPS)
 - C. Firewall
 - D. Honeypot
60. A company is considering implementing a system that allows employees to use their personal devices for work. Which of the following would be the biggest security concern?
- A. The devices may not be compatible with the company's software
 - B. The devices may not have the same level of security as company-owned devices
 - C. The devices may use more bandwidth than company-owned devices
 - D. The company may have to provide technical support for the devices
61. Which of the following is a characteristic of a symmetric encryption system?
- A. It uses different keys for encryption and decryption
 - B. It is slower than asymmetric encryption
 - C. It is less secure than asymmetric encryption
 - D. It uses the same key for encryption and decryption
62. Which of the following is the most secure method for storing passwords?
- A. Cleartext
 - B. Encrypted
 - C. Hashed
 - D. Salted and hashed
63. A security analyst is performing a vulnerability scan on a company's network. The scan reveals several open ports on a server. What is the risk associated with this finding?
- A. The server may be vulnerable to attack
 - B. The server is likely infected with malware
 - C. The server is likely part of a botnet
 - D. The server is likely being used for data exfiltration
64. A company has implemented a policy that requires all users to change their passwords every 90 days. What type of security control is this?
- A. Technical control
 - B. Physical control
 - C. Administrative control
 - D. Detective control

65. Which of the following is a type of social engineering attack where the attacker pretends to be a trusted entity to trick the victim into providing sensitive information?
- A. Spoofing
 - B. Phishing
 - C. DDoS attack
 - D. SQL injection
66. Which of the following is a security advantage of a network-based Intrusion Prevention System (NIPS) over a host-based Intrusion Prevention System (HIPS)?
- A. It can monitor all traffic on the network
 - B. It can prevent malware infections on individual hosts
 - C. It requires less computational resources to operate
 - D. It can detect attacks coming from inside the network
67. A company has implemented a system that requires users to swipe a card and enter a PIN to access the building. What type of authentication method is this?
- A. Single-factor authentication
 - B. Two-factor authentication
 - C. Multifactor authentication
 - D. Biometric authentication
68. A company is considering implementing a system that can detect attacks and alert security staff, but cannot take action to prevent the attacks. What type of system is this?
- A. Firewall
 - B. Intrusion Prevention System (IPS)
 - C. Intrusion Detection System (IDS)
 - D. Security Information and Event Management (SIEM) system
69. What is the primary purpose of a digital signature?
- A. To encrypt data
 - B. To verify the integrity of data
 - C. To authenticate users
 - D. To compress data

70. A company wants to implement a system that allows employees to securely connect to the corporate network from remote locations. What type of system should they implement?
- A. Firewall
 - B. Intrusion Detection System (IDS)
 - C. Virtual Private Network (VPN)
 - D. Network Access Control (NAC) system
71. A user reports that they have received an email from their bank asking them to click on a link to update their password. The user is suspicious because the email contains several spelling mistakes. What type of attack is this likely an example of?
- A. Vishing
 - B. Phishing
 - C. Whaling
 - D. Smishing
72. A company is implementing a new security policy that will require all employees to use a unique, complex password for each system they access. What type of attack is this policy designed to prevent?
- A. DDoS attack
 - B. Phishing attack
 - C. Man-in-the-Middle attack
 - D. Password cracking attack
73. A company is considering implementing a system that can provide secure remote access to the corporate network. Which of the following would be the best choice?
- A. Network Access Control (NAC) system
 - B. Intrusion Detection System (IDS)
 - C. Virtual Private Network (VPN)
 - D. Firewall
74. A security analyst is conducting a penetration test and has gained access to a system. The analyst now wants to escalate their privileges to gain more control over the system. What technique should the analyst use?
- A. Social engineering
 - B. Phishing
 - C. Privilege escalation
 - D. Spoofing

75. A company is implementing a new security policy that will require all data to be encrypted while it is in transit. What type of encryption should they use?
- A. Symmetric encryption
 - B. Asymmetric encryption
 - C. Hashing
 - D. Digital signatures
76. A company is considering implementing a system that uses artificial intelligence to detect and respond to security threats in real-time. What type of system is this?
- A. Intrusion Detection System (IDS)
 - B. Intrusion Prevention System (IPS)
 - C. Security Information and Event Management (SIEM) system
 - D. User and Entity Behavior Analytics (UEBA) system
77. A security analyst has detected a number of failed login attempts on a server. The login attempts are coming from multiple IP addresses and appear to be automated. What type of attack is most likely being attempted?
- A. Brute force attack
 - B. DDoS attack
 - C. Man-in-the-middle attack
 - D. SQL injection attack
78. A company wants to implement a system that can protect against zero-day attacks. What type of system should they implement?
- A. Intrusion Detection System (IDS)
 - B. Intrusion Prevention System (IPS)
 - C. Firewall
 - D. Antivirus software
79. A company has implemented a policy that requires all users to have a unique username and strong password. What type of security control is this?
- A. Technical control
 - B. Physical control
 - C. Administrative control
 - D. Detective control

80. Which of the following is a characteristic of a packet-filtering firewall?
- A. It can filter traffic based on the state of the connection
 - B. It cannot filter traffic based on IP addresses
 - C. It can filter traffic based on the content of the packets
 - D. It can filter traffic based on IP addresses and port numbers
81. Which of the following is a type of malware that can replicate itself and spread to other systems?
- A. Trojan
 - B. Virus
 - C. Spyware
 - D. Adware
82. A company is implementing a system that can provide secure remote access to the corporate network. The system uses a protocol that encrypts both the authentication process and the data in transit. What protocol is this?
- A. SSL
 - B. IPSec
 - C. SSH
 - D. TLS
83. A security analyst is considering implementing a system that uses behavior-based monitoring to detect threats. What type of system is this?
- A. Intrusion Detection System (IDS)
 - B. Intrusion Prevention System (IPS)
 - C. Antivirus software
 - D. Security Information and Event Management (SIEM) system
84. A company is considering outsourcing its IT services to a cloud provider. Which type of cloud deployment model would provide the company with its own private cloud infrastructure, but have it managed by the third-party provider?
- A. Public cloud
 - B. Private cloud
 - C. Hybrid cloud
 - D. Community cloud

85. Which of the following is a characteristic of a Public Key Infrastructure (PKI)?
- A. It uses symmetric encryption
 - B. It uses asymmetric encryption
 - C. It uses hashing
 - D. It uses digital signatures
86. A company is implementing a new security policy that will require all users to use a smart card and a fingerprint to access the corporate network. What type of authentication method is this?
- A. Single-factor authentication
 - B. Two-factor authentication
 - C. Multifactor authentication
 - D. Biometric authentication
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 - D. User and Entity Behavior Analytics (UEBA) system
88. Which of the following is a characteristic of ransomware?
- A. It replicates itself and spreads to other systems
 - B. It encrypts the victim's files and demands a ransom for the decryption key
 - C. It logs the victim's keystrokes to steal sensitive information
 - D. It displays unwanted advertisements on the victim's computer
89. A company is considering implementing a Bring Your Own Device (BYOD) policy. What is the primary security concern associated with this type of policy?
- A. The devices may not have the same level of security as company-owned devices
 - B. The company may have to provide technical support for the devices
 - C. The devices may not be compatible with the company's software
 - D. The devices may use more bandwidth than company-owned devices
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91. A company is considering implementing a system that can provide secure remote access to the corporate network. Which of the following would be the best choice?
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 - C. Virtual Private Network (VPN)
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92. A security analyst is conducting a penetration test and has gained access to a system. The analyst now wants to escalate their privileges to gain more control over the system. What technique should the analyst use?
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97. Which of the following is a vulnerability in the WEP protocol that makes it insecure?
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 - C. It is susceptible to man-in-the-middle attacks
 - D. It does not support two-factor authentication
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100. Which of the following is a type of security control that is implemented after a security incident has occurred?
- A. Preventive control
 - B. Detective control
 - C. Corrective control
 - D. Deterrent control
101. An IT professional is tasked with securing the company's data at rest. What would be an appropriate solution?
- A. Implement a firewall
 - B. Enforce strong password policy
 - C. Install antivirus software
 - D. Use full-disk encryption

102. A company is moving to a new office location. The CISO wants to ensure that the network wiring closets are secure. What type of security control should be implemented?
- A. Technical control
 - B. Physical control
 - C. Administrative control
 - D. Deterrent control
103. A penetration tester is trying to capture the NTLM hashes traversing the network. Which attack is being performed?
- A. Man-in-the-middle attack
 - B. Replay attack
 - C. Pass the hash attack
 - D. Rainbow table attack
104. What is the primary purpose of a Certificate Authority (CA) in a Public Key Infrastructure (PKI)?
- A. To distribute private keys to users
 - B. To verify the identity of users
 - C. To store certificates and keys
 - D. To issue and revoke digital certificates
105. A company wants to prevent sensitive information from being exfiltrated through email. Which technology should they implement?
- A. Network Access Control (NAC)
 - B. Intrusion Prevention System (IPS)
 - C. Data Loss Prevention (DLP)
 - D. Virtual Private Network (VPN)
106. A security analyst is investigating a security incident and finds evidence that an attacker has been attempting to guess the root password. What type of attack is this?
- A. Phishing
 - B. Brute force
 - C. Spoofing
 - D. Replay
107. Which security concept ensures that data is only viewable by authorized users?
- A. Confidentiality
 - B. Integrity
 - C. Availability
 - D. Non-repudiation

108. An organization is implementing a new security policy that requires users to use their fingerprint and a smart card to login. What type of authentication is this?
- A. Single-factor authentication
 - B. Two-factor authentication
 - C. Multifactor authentication
 - D. Biometric authentication
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 - B. Asymmetric encryption
 - C. Hashing
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117. Which of the following is the primary purpose of using a hash function in cybersecurity?
- A. To encrypt data
 - B. To verify data integrity
 - C. To compress data
 - D. To authenticate users
118. A security analyst is investigating an incident where an unauthorized user gained access to the network. The analyst suspects that the user was able to capture the network traffic and use it to gain access. What type of attack does this describe?
- A. Replay attack
 - B. Brute force attack
 - C. Man-in-the-middle attack
 - D. Phishing attack

119. A company wants to ensure that its wireless network is secure. Which of the following is a security standard for wireless networks that uses strong encryption and authentication?
- A. WEP
 - B. WPA
 - C. WPA2
 - D. WPA3
120. A company has implemented a security policy that requires all employees to change their password every 90 days. This is an example of what type of security control?
- A. Physical control
 - B. Technical control
 - C. Administrative control
 - D. Deterrent control
121. A company has decided to implement a security policy that requires users to authenticate using something they know and something they have. What type of authentication is this?
- A. Single-factor authentication
 - B. Two-factor authentication
 - C. Multifactor authentication
 - D. Biometric authentication
122. A security analyst has discovered a vulnerability in the company's network. The vulnerability allows an attacker to execute arbitrary code remotely. What type of vulnerability is this?
- A. Buffer overflow
 - B. SQL Injection
 - C. Cross-Site Scripting
 - D. Cross-Site Request Forgery
123. A company is implementing a new security policy that will require all data to be encrypted while it is in transit. What type of encryption should they use?
- A. Symmetric encryption
 - B. Asymmetric encryption
 - C. Hashing
 - D. Digital signatures

124. A company is considering implementing a system that uses artificial intelligence to detect and respond to security threats in real-time. What type of system is this?
- A. Intrusion Detection System (IDS)
 - B. Intrusion Prevention System (IPS)
 - C. Security Information and Event Management (SIEM) system
 - D. User and Entity Behavior Analytics (UEBA) system
125. A company is considering outsourcing its IT services to a cloud provider. Which type of cloud deployment model would provide the company with its own private cloud infrastructure, but have it managed by the third-party provider?
- A. Public cloud
 - B. Private cloud
 - C. Hybrid cloud
 - D. Community cloud
126. A security analyst is conducting a penetration test and has gained access to a system. The analyst now wants to maintain their access for future exploitation. What technique should the analyst use?
- A. Social engineering
 - B. Phishing
 - C. Creating a backdoor
 - D. Spoofing
127. A company wants to ensure that the software they develop is free of any security vulnerabilities. Which of the following would be the best approach to achieve this?
- A. Implement secure coding practices
 - B. Use antivirus software
 - C. Use a firewall
 - D. Encrypt all data in transit
128. A company is implementing a new security policy that requires all data to be encrypted while at rest. What type of encryption should they use?
- A. Symmetric encryption
 - B. Asymmetric encryption
 - C. Hashing
 - D. Digital signatures

129. Which of the following is a security concern associated with cloud computing?
- A. The company will lose control over its data
 - B. The company will have to purchase new hardware
 - C. The company will have to hire more IT staff
 - D. The company will have to implement a new network infrastructure
130. A company has implemented a policy that requires users to authenticate using a smart card and a PIN. What type of authentication is this?
- A. Single-factor authentication
 - B. Two-factor authentication
 - C. Multifactor authentication
 - D. Biometric authentication
131. A company is considering implementing a new system that can provide secure remote access to the corporate network. Which of the following would be the best choice?
- A. Network Access Control (NAC) system
 - B. Intrusion Detection System (IDS)
 - C. Virtual Private Network (VPN)
 - D. Firewall
132. A security analyst is investigating an incident where an unauthorized user gained access to the network. The analyst suspects that the user was able to capture the network traffic and use it to gain access. What type of attack does this describe?
- A. Replay attack
 - B. Brute force attack
 - C. Man-in-the-middle attack
 - D. Phishing attack
133. A company is considering implementing a system that uses artificial intelligence to detect and respond to security threats in real-time. What type of system is this?
- A. Intrusion Detection System (IDS)
 - B. Intrusion Prevention System (IPS)
 - C. Security Information and Event Management (SIEM) system
 - D. User and Entity Behavior Analytics (UEBA) system

134. A company's web server has been defaced by a hacker. What type of attack has likely occurred?
- A. DDoS attack
 - B. Phishing attack
 - C. Web application attack
 - D. Man-in-the-Middle attack
135. A company wants to implement an authentication system that uses a centralized server to authenticate users. Which of the following would be the best choice?
- A. TACACS+
 - B. CHAP
 - C. MS-CHAP
 - D. PAP
136. A penetration tester is attempting to compromise a system by exploiting a vulnerability that allows unauthorized access to a system's memory. What type of attack is this?
- A. SQL Injection
 - B. Buffer overflow
 - C. Cross-site scripting
 - D. Denial of service
137. A company wants to ensure that the data it transmits over the internet is protected from eavesdropping. Which of the following should it implement?
- A. A firewall
 - B. Antivirus software
 - C. An intrusion detection system
 - D. Transport Layer Security (TLS)
138. Which of the following is a type of malware that encrypts a user's files and demands payment to decrypt them?
- A. Worm
 - B. Trojan
 - C. Spyware
 - D. Ransomware
139. A security analyst is conducting a vulnerability scan on a company's network. The scan returns a list of open ports on various systems. What type of information is this?
- A. User information
 - B. System configuration information
 - C. Network topology information
 - D. Asset inventory information

140. A security analyst is investigating a security incident and has determined that an attacker gained access to a system by exploiting a vulnerability in the operating system. What type of patch should be applied to prevent future incidents?
- A. Firmware update
 - B. Security patch
 - C. Software update
 - D. Driver update
141. What is the primary purpose of a firewall in a company's network infrastructure?
- A. To protect the network from malware
 - B. To monitor network traffic
 - C. To control network traffic
 - D. To encrypt network traffic
142. A company is considering implementing a new security policy that requires all data to be encrypted while it is in transit. What type of encryption should they use?
- A. Symmetric encryption
 - B. Asymmetric encryption
 - C. Hashing
 - D. Digital signatures
143. A security analyst has discovered a vulnerability in a web application that allows an attacker to inject malicious code into the application. What type of vulnerability is this?
- A. SQL Injection
 - B. Cross-Site Scripting
 - C. Buffer Overflow
 - D. Cross-Site Request Forgery
144. An IT manager is concerned about the possibility of employees inadvertently installing malicious software on their company-issued laptops. Which of the following controls would BEST mitigate this risk?
- A. Implementing a strict password policy
 - B. Disabling USB ports on all company-issued laptops
 - C. Installing antivirus software on all company-issued laptops
 - D. Implementing a software whitelist

145. A company's web server has been defaced by a hacker. What type of attack has likely occurred?
- A. DDoS attack
 - B. Phishing attack
 - C. Web application attack
 - D. Man-in-the-Middle attack
146. A security analyst is investigating an incident where an unauthorized user gained access to the network. The analyst suspects that the user was able to capture the network traffic and use it to gain access. What type of attack does this describe?
- A. Replay attack
 - B. Brute force attack
 - C. Man-in-the-middle attack
 - D. Phishing attack
147. A company has implemented a policy that requires users to authenticate using a smart card and a PIN. What type of authentication is this?
- A. Single-factor authentication
 - B. Two-factor authentication
 - C. Multifactor authentication
 - D. Biometric authentication
148. A company is considering implementing a new system that can provide secure remote access to the corporate network. Which of the following would be the best choice?
- A. Network Access Control (NAC) system
 - B. Intrusion Detection System (IDS)
 - C. Virtual Private Network (VPN)
 - D. Firewall
149. A company is implementing a new security policy that requires all data to be encrypted while it is in transit. What type of encryption should they use?
- A. Symmetric encryption
 - B. Asymmetric encryption
 - C. Hashing
 - D. Digital signatures

150. A company is considering implementing a system that uses artificial intelligence to detect and respond to security threats in real-time. What type of system is this?
- A. Intrusion Detection System (IDS)
 - B. Intrusion Prevention System (IPS)
 - C. Security Information and Event Management (SIEM) system
 - D. User and Entity Behavior Analytics (UEBA) system
151. A security analyst is conducting a penetration test and has gained access to a system. The analyst now wants to escalate their privileges to gain more control over the system. What technique should the analyst use?
- A. Social engineering
 - B. Phishing
 - C. Privilege escalation
 - D. Spoofing
152. A company has decided to implement a security policy that requires users to authenticate using something they know and something they have. What type of authentication is this?
- A. Single-factor authentication
 - B. Two-factor authentication
 - C. Multifactor authentication
 - D. Biometric authentication
153. A company is considering outsourcing its IT services to a cloud provider. Which type of cloud deployment model would provide the company with its own private cloud infrastructure, but have it managed by the third-party provider?
- A. Public cloud
 - B. Private cloud
 - C. Hybrid cloud
 - D. Community cloud
154. A penetration tester is attempting to compromise a system by exploiting a vulnerability that allows unauthorized access to a system's memory. What type of attack is this?
- A. SQL Injection
 - B. Buffer overflow
 - C. Cross-site scripting
 - D. Denial of service

155. A company wants to ensure that the data it transmits over the internet is protected from eavesdropping. Which of the following should it implement?
- A. A firewall
 - B. Antivirus software
 - C. An intrusion detection system
 - D. Transport Layer Security (TLS)
156. A security analyst is evaluating a web application for potential vulnerabilities. The analyst notices that the application doesn't properly validate input, which might allow an attacker to run malicious database commands. What type of vulnerability has the analyst discovered?
- A. Cross-Site Request Forgery (CSRF)
 - B. Cross-Site Scripting (XSS)
 - C. SQL Injection
 - D. Buffer Overflow
157. A company's Security Operations Center (SOC) has detected an unusual amount of traffic from multiple sources targeting a single server within the network. This traffic is consuming all of the server's resources, making it unavailable for legitimate users. Which type of attack is MOST likely taking place?
- A. Man-in-the-Middle Attack
 - B. Distributed Denial-of-Service Attack
 - C. Social Engineering Attack
 - D. Replay Attack
158. A company wants to allow employees to work remotely. However, they want to ensure that the data transmitted between the remote employees and the corporate network is secure. What technology should they implement?
- A. Intrusion Detection System (IDS)
 - B. Virtual Private Network (VPN)
 - C. Wireless Access Point (WAP)
 - D. Firewall
159. A security analyst receives an email from a trusted source asking for sensitive information. However, upon closer inspection, the analyst realizes that the email is actually from an attacker impersonating the trusted source. Which type of attack has the analyst experienced?
- A. Spear Phishing
 - B. Vishing
 - C. Pharming
 - D. Whaling

160. Which of the following describes a system or software vulnerability that is unknown to the vendor and therefore unpatched?
- A. Zero-Day Vulnerability
 - B. Known Vulnerability
 - C. Residual Risk
 - D. Threat Vector
161. A company wants to verify the integrity of a file downloaded from the internet. Which of the following should they use?
- A. Asymmetric encryption
 - B. Symmetric encryption
 - C. Hashing
 - D. Digital signature
162. A company is implementing a new web application. The security team is concerned about the potential for injection attacks. What type of security control should the team implement to mitigate this risk?
- A. Input validation
 - B. Intrusion Detection System (IDS)
 - C. Firewall
 - D. Antivirus software
163. A security analyst is performing a risk assessment. Which of the following is a calculation that the analyst should use to determine risk?
- A. $\text{Threat} \times \text{Vulnerability} \times \text{Impact}$
 - B. $\text{Threat} \times \text{Vulnerability} / \text{Impact}$
 - C. $\text{Threat} / \text{Vulnerability} \times \text{Impact}$
 - D. $\text{Threat} + \text{Vulnerability} + \text{Impact}$
164. An attacker has gained access to a network and is attempting to evade detection by blending in with normal network traffic. Which of the following BEST describes this type of attack?
- A. Trojan
 - B. Man-in-the-middle
 - C. Denial of Service
 - D. Covert channel

165. A company is considering implementing biometric authentication. Which of the following is a disadvantage of biometric authentication that the company should consider?
- A. It can be easily forgotten
 - B. It can be easily copied
 - C. It is not unique
 - D. It can change over time
166. A user receives an email claiming to be from their bank, asking them to click on a link and verify their account information. This is an example of which type of attack?
- A. Vishing
 - B. Phishing
 - C. Spoofing
 - D. Spamming
167. A security analyst is conducting a vulnerability assessment and determines that a system is at risk of attack because it has an unpatched operating system. What type of vulnerability is this?
- A. Configuration vulnerability
 - B. Software vulnerability
 - C. Hardware vulnerability
 - D. Network vulnerability
168. A company wants to ensure that only authorized devices can connect to their network. Which of the following would be the MOST effective way to achieve this?
- A. Implement a firewall
 - B. Enable MAC address filtering
 - C. Use strong passwords
 - D. Enable network encryption
169. A penetration tester has successfully exploited a system and now wants to ensure that they can continue to have access to the system, even if the vulnerability they exploited is patched. What should the penetration tester do?
- A. Install a rootkit
 - B. Conduct a denial of service attack
 - C. Use an encryption algorithm
 - D. Implement a firewall

170. A security analyst is reviewing a company's security policies and notices that there is no policy for regularly reviewing and updating user access rights. Which of the following principles of access control is the company failing to implement?
- A. Least privilege
 - B. Separation of duties
 - C. Mandatory access control
 - D. Access control list
171. A company has hired a new IT support technician. The technician needs to be able to reset user passwords, but should not have access to any other administrative functions. Which of the following principles of access control is being implemented?
- A. Least privilege
 - B. Separation of duties
 - C. Mandatory access control
 - D. Access control list
172. A security analyst is investigating an incident in which an attacker was able to capture network traffic and replay it to gain unauthorized access. Which of the following should the analyst recommend to prevent this type of attack in the future?
- A. Implement a firewall
 - B. Use strong passwords
 - C. Enable network encryption
 - D. Implement time-based one-time passwords
173. An organization wants to ensure that their systems remain operational even in the event of a system failure or disaster. Which of the following would be the BEST solution?
- A. Implement a firewall
 - B. Implement an intrusion detection system
 - C. Implement a backup and recovery solution
 - D. Implement an antivirus solution
174. A company is implementing a new security policy that requires users to change their passwords every 90 days. Which of the following types of attacks is this policy designed to mitigate?
- A. Brute force attacks
 - B. Man-in-the-middle attacks
 - C. Denial of service attacks
 - D. Replay attacks

175. A security analyst is reviewing the company's use of encryption and notices that data at rest is not being encrypted. Which of the following would be the BEST solution to address this issue?
- A. Implement full-disk encryption
 - B. Implement a firewall
 - C. Implement an intrusion detection system
 - D. Implement a virtual private network
176. A company wants to allow employees to use their personal devices for work, but is concerned about the security risks. Which of the following would be the BEST policy to implement?
- A. Acceptable use policy
 - B. Password policy
 - C. Bring your own device (BYOD) policy
 - D. User access policy
177. An attacker has installed a rogue wireless access point that mimics a legitimate access point to trick users into connecting to it. This is an example of which type of attack?
- A. Man-in-the-middle attack
 - B. Evil twin attack
 - C. Denial of service attack
 - D. Replay attack
178. A security analyst is reviewing network logs and notices multiple login attempts to a system from a foreign IP address. All of the login attempts use different usernames and passwords. Which type of attack is this MOST likely?
- A. Brute force attack
 - B. Phishing attack
 - C. Man-in-the-middle attack
 - D. Replay attack
179. A company wants to ensure that only authorized users can access the network. Which of the following is a common method for verifying the identity of users?
- A. Encryption
 - B. Authentication
 - C. Non-repudiation
 - D. Confidentiality

180. A security analyst is investigating an incident in which an attacker sent an email to a user with a malicious attachment. When the user opened the attachment, a program was installed that allowed the attacker to control the user's system. What type of malware does this describe?
- A. Worm
 - B. Trojan
 - C. Virus
 - D. Ransomware
181. A company is considering using a third-party cloud service provider to store its data. However, the company is concerned about the security of its data. Which of the following can be used to provide security assurance in cloud computing?
- A. Service level agreement (SLA)
 - B. Risk assessment
 - C. Data loss prevention (DLP)
 - D. Intrusion detection system (IDS)
182. Which of the following types of cryptographic algorithms provides both confidentiality and non-repudiation?
- A. Symmetric encryption
 - B. Asymmetric encryption
 - C. Hashing
 - D. Steganography
183. A security analyst is conducting a vulnerability scan and finds that a server is running an unnecessary service that could be exploited by an attacker. What type of vulnerability is this?
- A. Configuration vulnerability
 - B. Software vulnerability
 - C. Hardware vulnerability
 - D. Network vulnerability
184. A company wants to ensure that their wireless network is secure. Which of the following encryption methods is currently considered the MOST secure for wireless networks?
- A. WEP
 - B. WPA
 - C. WPA2
 - D. WPA3

185. An organization has implemented a system to allow employees to use their network username and password to log into multiple systems. This is an example of which of the following?
- A. Multifactor authentication
 - B. Single sign-on
 - C. Two-factor authentication
 - D. Biometric authentication
186. A security analyst is conducting a penetration test and is trying to exploit a known vulnerability in a web application. However, the analyst is unable to exploit the vulnerability because the system has been patched. What type of testing is the analyst conducting?
- A. White box testing
 - B. Black box testing
 - C. Grey box testing
 - D. Red team testing
187. A company wants to implement a system that can detect potential security threats and respond to them in real-time. Which of the following would be the BEST solution?
- A. Intrusion Detection System (IDS)
 - B. Intrusion Prevention System (IPS)
 - C. Firewall
 - D. Antivirus software
188. A company is planning to implement a new security policy that requires users to authenticate using something they have and something they know. What type of authentication is this?
- A. Single-factor authentication
 - B. Two-factor authentication
 - C. Multifactor authentication
 - D. Biometric authentication
189. A company wants to ensure that its data is protected from unauthorized access, even if a device is lost or stolen. Which of the following would be the BEST solution?
- A. Implement a firewall
 - B. Implement full disk encryption
 - C. Implement strong passwords
 - D. Implement an intrusion detection system

190. A company is planning to implement a new security policy that requires users to authenticate using a fingerprint scanner. This is an example of which of the following types of authentication?
- A. Something you know
 - B. Something you have
 - C. Something you are
 - D. Somewhere you are
191. A security analyst is reviewing the results of a vulnerability scan and notices that a server is missing a critical security patch. Which of the following should the analyst do FIRST?
- A. Apply the patch
 - B. Conduct a risk assessment
 - C. Notify management
 - D. Shutdown the server
192. A company wants to ensure that their data remains confidential, even while it is being transmitted over the internet. Which of the following would be the BEST solution?
- A. Use strong passwords
 - B. Implement a firewall
 - C. Implement a virtual private network (VPN)
 - D. Implement an intrusion detection system
193. A security analyst is reviewing network logs and notices a large number of requests to a particular website. The analyst suspects that this is a distributed denial of service (DDoS) attack. Which of the following would be the BEST course of action?
- A. Block all traffic to the website
 - B. Monitor the traffic to identify the source of the attack
 - C. Implement a firewall to protect the network
 - D. Notify law enforcement
194. A company is planning to implement a system that will detect and prevent unauthorized access to the network. Which of the following would be the BEST solution?
- A. Intrusion Detection System (IDS)
 - B. Intrusion Prevention System (IPS)
 - C. Firewall
 - D. Antivirus software

195. A company has implemented a system that requires users to swipe a card and enter a PIN to access a secure area. This is an example of which of the following types of authentication?
- A. Single-factor authentication
 - B. Two-factor authentication
 - C. Multifactor authentication
 - D. Biometric authentication
196. A security analyst is reviewing the company's security policies and notices that there is no policy for regularly testing the company's security controls. Which of the following should the analyst recommend?
- A. Implement a firewall
 - B. Conduct regular vulnerability assessments
 - C. Use strong passwords
 - D. Implement an intrusion detection system
197. A company wants to ensure that their data is protected, even if an attacker is able to gain physical access to their servers. Which of the following would be the BEST solution?
- A. Implement a firewall
 - B. Implement full disk encryption
 - C. Implement an intrusion detection system
 - D. Use strong passwords
198. A company is concerned about the potential for data leakage. Which of the following would be the BEST solution?
- A. Implement a firewall
 - B. Implement an intrusion detection system
 - C. Implement a data loss prevention (DLP) system
 - D. Use strong passwords
199. A security analyst is reviewing network logs and notices a large number of failed login attempts to a particular system. Which of the following types of attacks is this MOST likely?
- A. Brute force attack
 - B. Phishing attack
 - C. Denial of service attack
 - D. Man-in-the-middle attack

200. A company is implementing a new security policy that requires all employees to use a smart card and a PIN to log into their computers. This is an example of which of the following types of authentication?
- A. Single-factor authentication
 - B. Two-factor authentication
 - C. Multifactor authentication
 - D. Biometric authentication

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1. C	51. A	101.D	151.C
2. B	52. C	102.B	152.B
3. B	53. A	103.C	153.B
4. A	54. B	104.D	154.B
5. C	55. B	105.C	155.D
6. C	56. C	106.B	156.C
7. C	57. C	107.A	157.B
8. D	58. C	108.C	158.B
9. C	59. B	109.C	159.A
10. C	60. B	110.B	160.A
11. B	61. D	111.B	161.C
12. B	62. D	112.C	162.A
13. B	63. A	113.D	163.A
14. B	64. C	114.C	164.D
15. B	65. B	115.A	165.D
16. C	66. A	116.B	166.B
17. C	67. B	117.B	167.B
18. B	68. C	118.A	168.B
19. C	69. B	119.D	169.A
20. C	70. C	120.C	170.A
21. C	71. B	121.B	171.A
22. C	72. D	122.A	172.D
23. D	73. C	123.B	173.C
24. C	74. C	124.D	174.A
25. B	75. B	125.B	175.A
26. B	76. D	126.C	176.C
27. B	77. A	127.A	177.B
28. B	78. B	128.A	178.A
29. B	79. A	129.A	179.B
30. C	80. D	130.B	180.B
31. D	81. B	131.C	181.A
32. A	82. C	132.A	182.B
33. C	83. A	133.D	183.A
34. C	84. B	134.C	184.D
35. B	85. B	135.A	185.B
36. B	86. C	136.B	186.C
37. C	87. D	137.D	187.B
38. D	88. B	138.D	188.B
39. B	89. A	139.B	189.B
40. B	90. B	140.B	190.C
41. D	91. C	141.C	191.B
42. B	92. C	142.B	192.C
43. A	93. B	143.B	193.B
44. C	94. D	144.D	194.B
45. C	95. C	145.C	195.B
46. A	96. A	146.A	196.B
47. C	97. B	147.B	197.B
48. A	98. B	148.C	198.C
49. B	99. C	149.B	199.A
50. D	100.C	150. D	200.B